ODSA_OG_regions

🧠 ODSA_README
🧠 ODSA-AIM_and_METHOD

CONSIDERATIONS

1. Due to the fundamental sampling limitation of the available radar data: the method needs multi-azimuth coverage within a region to constrain the $\cos^2\theta$ model, and single flight lines or narrow angular distributions can't provide that.
2. It might be worth breaking datasets like MS by trajectory orientation or sub-region to see if the wavelength distributions cluster differently in the alpine vs selective erosion zones.

SUMMARY

Landscape classification

Region	IFPA (match for radar data)	ODSA	v18 - match?
TEST_Aurora_SB	Low relief / Selective erosion	Hard bed	**NO. insufficient spatial sampling**
ROSS_ICECAP	Low relief / Selective erosion	Heterogeneous	**insufficient spatial sampling**
PEL_CHA2	Selective erosion / **Alpine** (subglacial)	**Alpine**	**Partial: insufficient spatial sampling**
Moller_Stream	Selective erosion / Alpine (subglacial) / Alpine (subaereal)	Isotropic (Alpine?)	**insufficient spatial sampling**

Region	IFPA (match for radar data)	ODSA	v18 - match?
Thwaites_BAS	Alpine (subglacial) / Selective erosion (relict)	Alpine	**insufficient spatial sampling**
~~Thwaites_CR~~	Alpine (subglacial) / Selective erosion (relict)	Selective erosion (relict) 🚩 β is unreliable	**insufficient spatial sampling**
DML_AniRES	Selective erosion (ice streams) / invalid data	Alpine?	**NO**

Wavelength analysis

Region	Count	Mean (m)	Range	Candidates (>L/2)
TEST_Aurora_SB	158	343	51-7399	none
ROSS_ICECAP	354	328	42-14937	2 (7.5--9.0)km
PEL_CHA2	170	504	43-32645	none
Moller_Stream	6319	481	30-33970	10 (6.3--33.0)km
Thwaites_BAS	1792	814	30-28845	6 (6.0--24.7)km
~~Thwaites_CR~~	17548	1347	106-47828	33 (7.4--48.8)km
DML_AniRES	1587	659	68-31050	3 (6.4--34.1)km

Elevation analysis

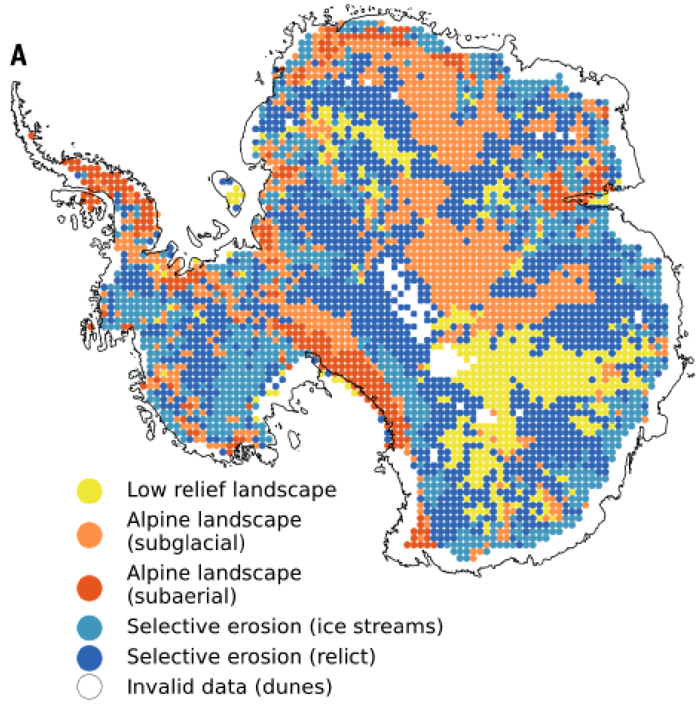
Region	Vertical relief (m)	Bed Elevation range (m)	IFPA includes "low-relief"?	ODSA
TEST_Aurora_SB	2003	-3479 to +1846	**yes**	deep basins AND high points
ROSS_ICECAP	1269	-1334 to +2746	**yes**	deep basins AND

Region	Vertical relief (m)	Bed Elevation range (m)	IFPA includes "low-relief"?	ODSA
				high points
PEL_CHA2	2194	-1073 to +2912	no	deep basins AND high points
Moller_Stream	1837 (mean, range: 488--2921)	-2115 to +1818	no	deep basins AND high points
Thwaites_BAS	665 (mean, range: 16--1538)	-1270 to +2380	no	deep basins AND high points
~~**Thwaites_CR**~~	178 (mean, range: 95--328)	-1641 to -896	no	deep basin
DML_AniRES	448 (mean, range: 153--918)	-952 to +534	no	deep basin

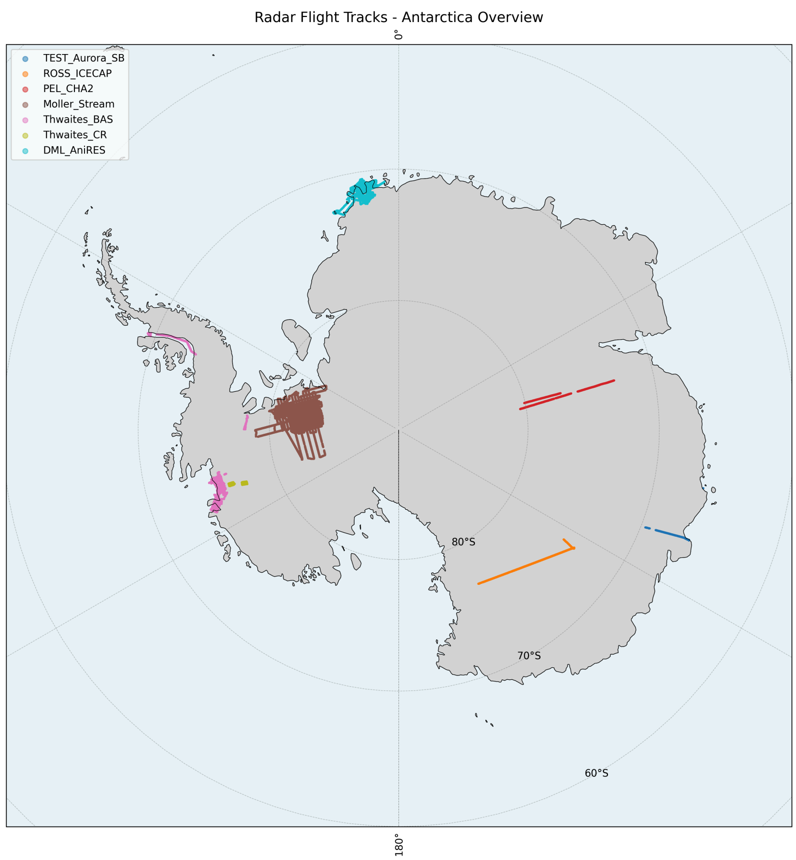
Problems

1. Data **tracks** present problems when they **are too long** and **cover very distinct landscapes**. (Spatial heterogeneity within regions if β varies spatially (not just directionally), pooling all windows together dilutes any anisotropic signal)
2. Window-level β has high intrinsic scatter — the single-pass `polyfit` per window is noisy; the signal-to-noise ratio may be too low for the cos² model to emerge
3. Flow direction uncertainty — the REMA-derived flow vectors introduce their own error, smearing the angular relationship

Ockenden Fig. 4. IFPA Geomorphological classification of Antarctica's subglacial landscape



Selected regions: Radar tracks



RESULTS

Aurora Subglacial Basin (ASB)

Hypothesis 1

High anisotropy ($\beta_{\parallel} > \beta_{\perp}$), **Selective erosion (ice streams)**

Terminal

```
=== TEST_Aurora_SB SUMMARY ===
------------------------------------------------------------
  RESULTS SUMMARY (1 trajectories aggregated)
------------------------------------------------------------
VERTICAL RELIEF (Max-Min):
  -> 2002.5m (Single Value)
ABSOLUTE BED ELEVATION:
  -> Min: -3478.6m | Max: 1845.5m  (across all trajectories)
AVG SEGMENT LENGTH:
  -> 109622.9m (Single Value)
POWER LAW EXPONENT (Beta):
  -> Mean: 1.6 | Range: [1.3, 2.2]
HURST EXPONENT:
  -> Mean: 0.3 | Range: [0.2, 0.6]
MEAN ICE THICKNESS:
  -> 1974.1m (Single Value)
............................................................
CONFIRMED WAVELENGTHS (Physically valid < L/2):
  -> Count: 158
  -> Mean: 343.3m | Range: [51.0, 7398.8]m
............................................................
LARGEST LOCAL STRUCTURES (50km Window):
  -> Relief: 1555.5m at km 170.7
............................................................
============================================================
Exported 27 window rows and 3 segment rows
```

Observations

IFPA categorised the region containing these radar tracks as **Selective erosion**
ODSA results indicate this region has a mean β value of 1.6 with a range: [1.3, 2.2].

Putting the result closest to the **hard bed** ("Brownian": complex and rough) class.

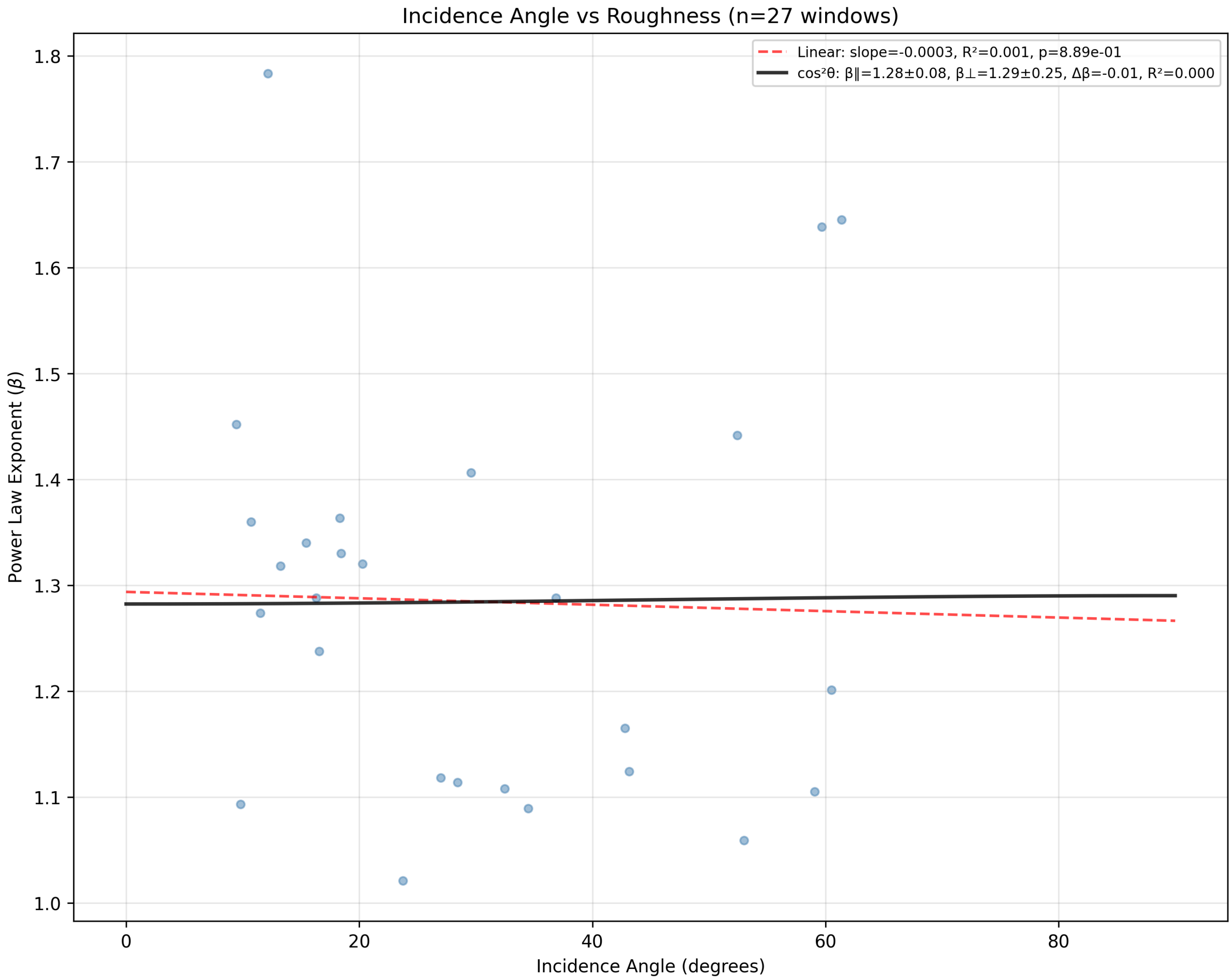
The raw max-minus-min elevation value for the vertical relief of 2002.5 m (single trajectory)

The height difference between the highest and lowest points available in the data for this area are Min: -3478.6 m | Max: 1845.5 m (across the single available trajectory).

The mean confirmed wavelength of 343.3 m with the range going up to ~8 km indicates the **roughness is dominated by structures at roughly the sub-km-scale**. These are closely-spaced features, which are consistent with a hard-bed environment where the bedrock itself (crystalline) creates high-frequency roughness. The low wavelength mean also fits with β=1.6 -- a flatter power spectrum means more relative power at short wavelengths, so the peak-finding identifies many small features. The lack of any large-scale periodic structures suggests no major valley/ridge systems within this track.

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=-0.0003, R^2=0.001, p=8.89e-01

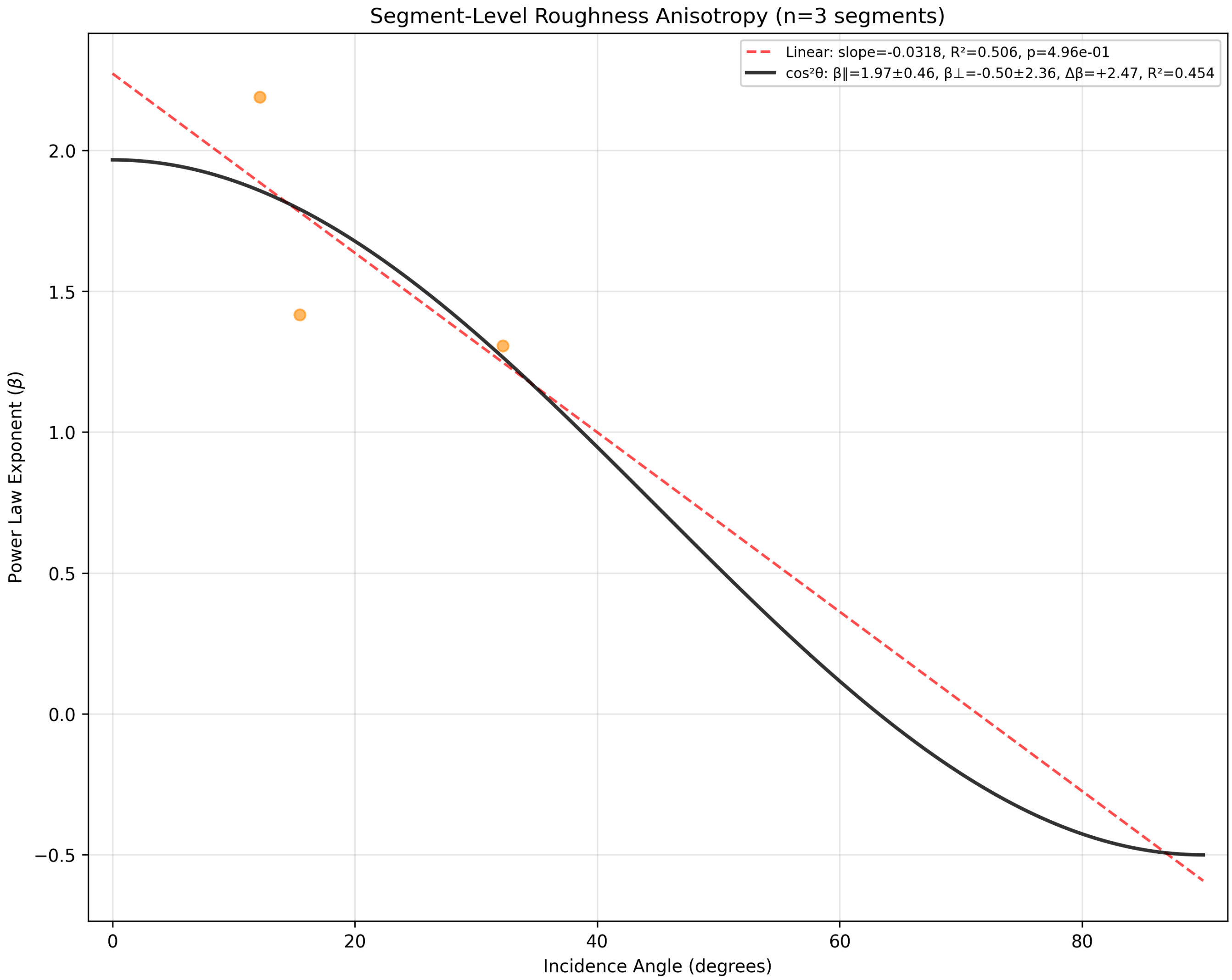
At p=0.889 this result is non-significant (there's a 89% chance you'd see this slope (or steeper) from pure noise). This means that the slope is essentially zero.

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\|} - \beta_{\perp} = -0.008$, which provides support for the **Alpine landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\|}^2+\sigma_{\perp}^2}} = \frac{0.008}{\sqrt{0.08^2+0.25^2}} \approx \frac{0.008}{0.069} \approx 0.262$

This is below 1.96 (95% confidence interval), which means that the result is not significant. Since the dataset is small and the R^2=0.0001 implies that the model explains only 0.01% of variance.This means that **essentially no directional dependence is detected by the model.** The result is uninterpretable at this scale

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=-0.0318, R^2=0.506, p=4.96e-01
At p=0.496 this result is non-significant (there's a 50% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\|} - \beta_{\perp} = +2.467$, which provides support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\|}^2+\sigma_{\perp}^2}} = \frac{2.467}{\sqrt{0.463^2+2.365^2}} \approx \frac{2.467}{2.4098} \approx 1.02$

This is below 1.96 (95% confidence interval), which means that the result is not significant. R^2=0.4537 implies that the model explains 45% of variance. However the The dataset is too small to draw any meaningful conclusion from this analysis.

Conclusion:

Insufficient data: 1 trajectory, 3 segments, 27 windows. The anisotropy test has essentially no power here.

Ross

> ✎ **Hypothesis**
>
> **Low relief**, **Alpine Selective erosion (relict)**

Terminal

```
=== ROSS_ICECAP SUMMARY ===
------------------------------------------------------------
  RESULTS SUMMARY (1 trajectories aggregated)
------------------------------------------------------------
VERTICAL RELIEF (Max-Min):
  -> 1269.0m (Single Value)
ABSOLUTE BED ELEVATION:
  -> Min: -1333.6m | Max: 2745.9m  (across all trajectories)
AVG SEGMENT LENGTH:
  -> 159751.5m (Single Value)
POWER LAW EXPONENT (Beta):
  -> Mean: 2.5 | Range: [1.8, 3.6]
HURST EXPONENT:
  -> Mean: 0.7 | Range: [0.4, 1.3]
MEAN ICE THICKNESS:
  -> 2380.3m (Single Value)
............................................................
CONFIRMED WAVELENGTHS (Physically valid < L/2):
  -> Count: 354
  -> Mean: 328.4m | Range: [42.0, 14936.7]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
  -> Count: 2
  -> Range: [7507m, 8981m]
............................................................
LARGEST LOCAL STRUCTURES (50km Window):
  -> Relief: 1822.9m at km 870.1
```

```
...............................................................
===============================================================
Exported 81 window rows and 6 segment rows
```

Observations

IFPA categorised the region containing these radar tracks as a highly heterogenous region with at least 3 distinct classifications for landscape. The ODSA results indicate this region has a mean β value of 2.5 with a range: [1.8, 3.6]: Putting the result in a combination landscape between hard bedrock and soft sediment.

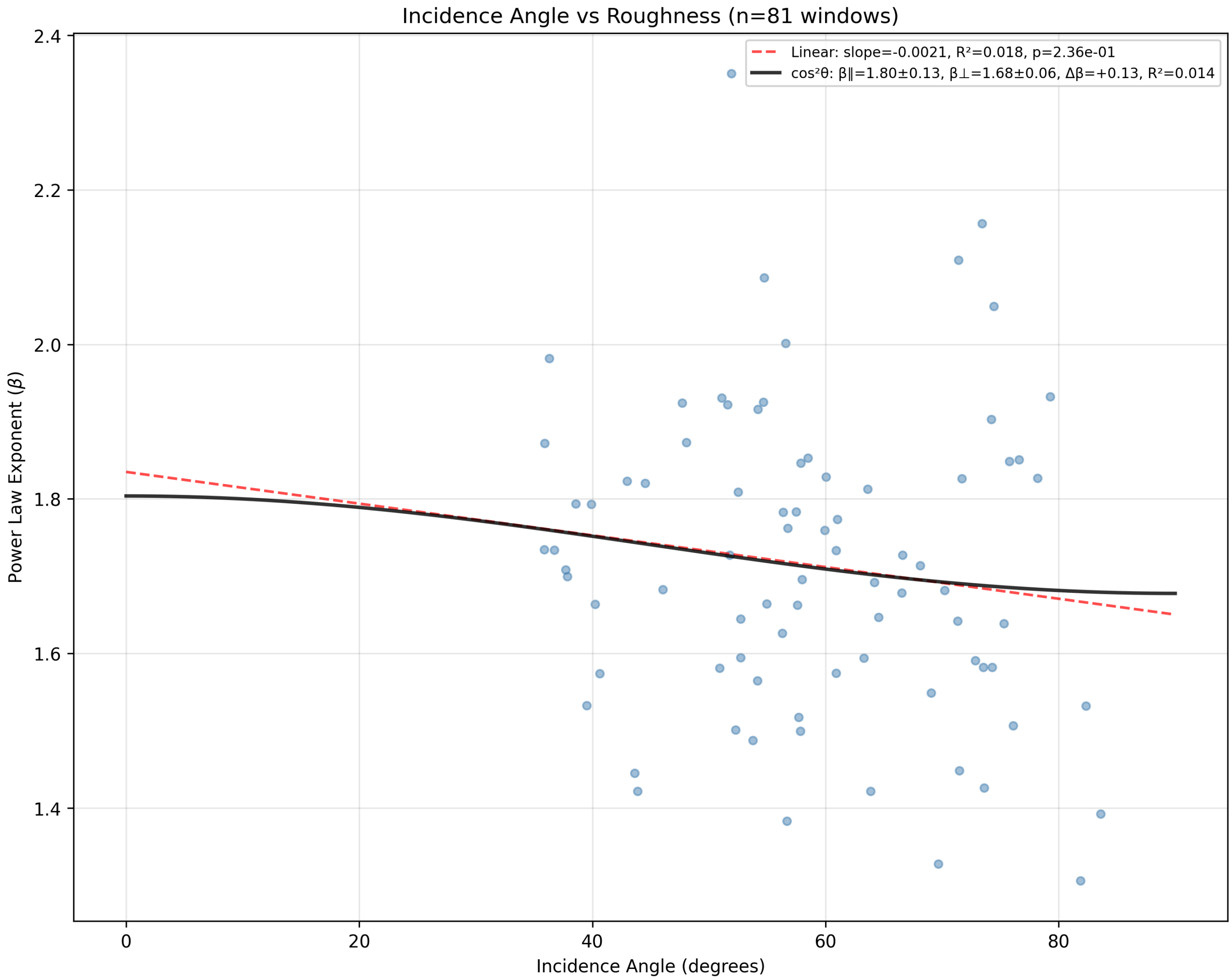
The raw max-minus-min elevation value for the vertical relief of 1269 m (single trajectory)

The height difference between the highest and lowest points available in the data for this area are Min: -1333.6 m | Max: 2745.9 m (across the single available trajectory).

The mean confirmed wavelength of 328.4 m with the range going up to ~15 km indicates the roughness is dominated by structures at roughly the sub-km-scale.
But the range extension is hinting at large-scale structuring that the 50 km window can't fully resolve. Given that IFPA classifies this region as heterogeneous (low relief + selective erosion + alpine), those ~9 km candidates could correspond to broad
subglacial valley spacing. The $\beta = 2.5$ indicates the power spectrum is steep and long
wavelengths carry most of the power, yet the confirmed peaks are mostly small. This could mean the large-scale topography is more of a monotonic trend (removed by detrending) rather than periodic structure.

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=-0.0021, R^2=0.018, p=2.36e-01

At p=0.236 this result is non-significant.

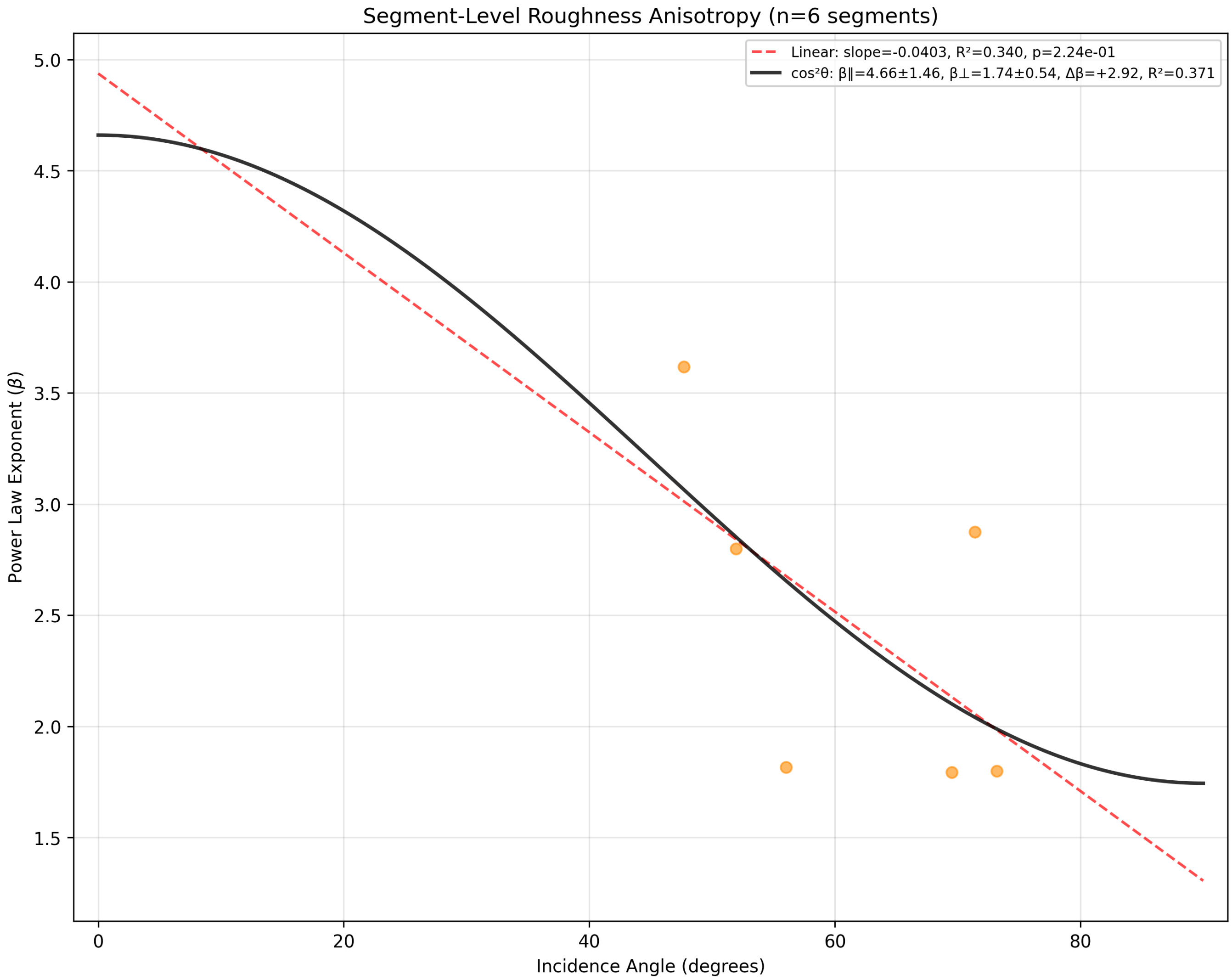
The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.13$, which provides support for the **Selective erosion** class.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2+\sigma_{\perp}^2}} = \frac{0.126}{\sqrt{0.128^2+0.065^2}} \approx \frac{0.126}{0.021} \approx 0.144$

This is below 1.96 (95% confidence interval), which means that the result is not significant.

The R^2=0.0142 value implies that the model explains only 1.42% of variance. This means that **essentially no directional dependence is detected by the model.**

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=-0.0403, R^2=0.340, p=2.24e-01
At p=0.224 this result is non-significant (there's a 22% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +2.915$, which provides support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2+\sigma_{\perp}^2}} = \frac{2.915}{\sqrt{1.465^2+0.536^2}} \approx \frac{2.915}{1.559} \approx 1.869$
This is below 1.96 (95% confidence interval), which means that the result is not significant. R^2=0.3710 implies that the model explains 37% of variance. However the The dataset is too small to draw any meaningful conclusion from this analysis.

Conclusion:

Model shows no systematic dependence of roughness on flow orientation.
Landscape appears to be too diverse within this track.
Data is insufficient to draw any conclusions: 1 trajectory, 6 segments, and 81 windows, the β range [1.8, 3.6] spanning nearly the entire classification spectrum suggests heterogeneous terrain within the single track.

Princess Elizabeth Land (PEL)

> **Hypothesis**
>
> Reversed anisotropy ($\beta_{\|} < \beta_{\perp}$) OR **isotropy** ($\beta_{\|} = \beta_{\perp}$).
> **High relief** and $\beta \in (1.5, 2.0)$
> Selective erosion / Alpine (subglacial)

Terminal

```
=== PEL_CHA2 SUMMARY ===
------------------------------------------------------------
  RESULTS SUMMARY (1 trajectories aggregated)
------------------------------------------------------------
VERTICAL RELIEF (Max-Min):
  -> 2194.1m (Single Value)
ABSOLUTE BED ELEVATION:
  -> Min: -1073.1m | Max: 2912.3m  (across all trajectories)
AVG SEGMENT LENGTH:
  -> 250615.9m (Single Value)
POWER LAW EXPONENT (Beta):
  -> Mean: 2.3 | Range: [2.0, 2.5]
HURST EXPONENT:
  -> Mean: 0.6 | Range: [0.5, 0.7]
MEAN ICE THICKNESS:
  -> 1696.8m (Single Value)
............................................................
CONFIRMED WAVELENGTHS (Physically valid < L/2):
  -> Count: 170
  -> Mean: 503.6m | Range: [42.6, 32645.3]m
............................................................
```

```
LARGEST LOCAL STRUCTURES (50km Window):
  -> Relief: 1817.3m at km 628.7
............................................................
============================================================
Exported 61 window rows and 3 segment rows
```

Observations

IFPA categorised the region containing these radar tracks as **Selective erosion / Alpine (subglacial)**. The ODSA results indicate this region has a mean β value of 2.3 with a range: [2.0, 2.5]. Putting the result between the **hard bed** and the **soft bed** categories.

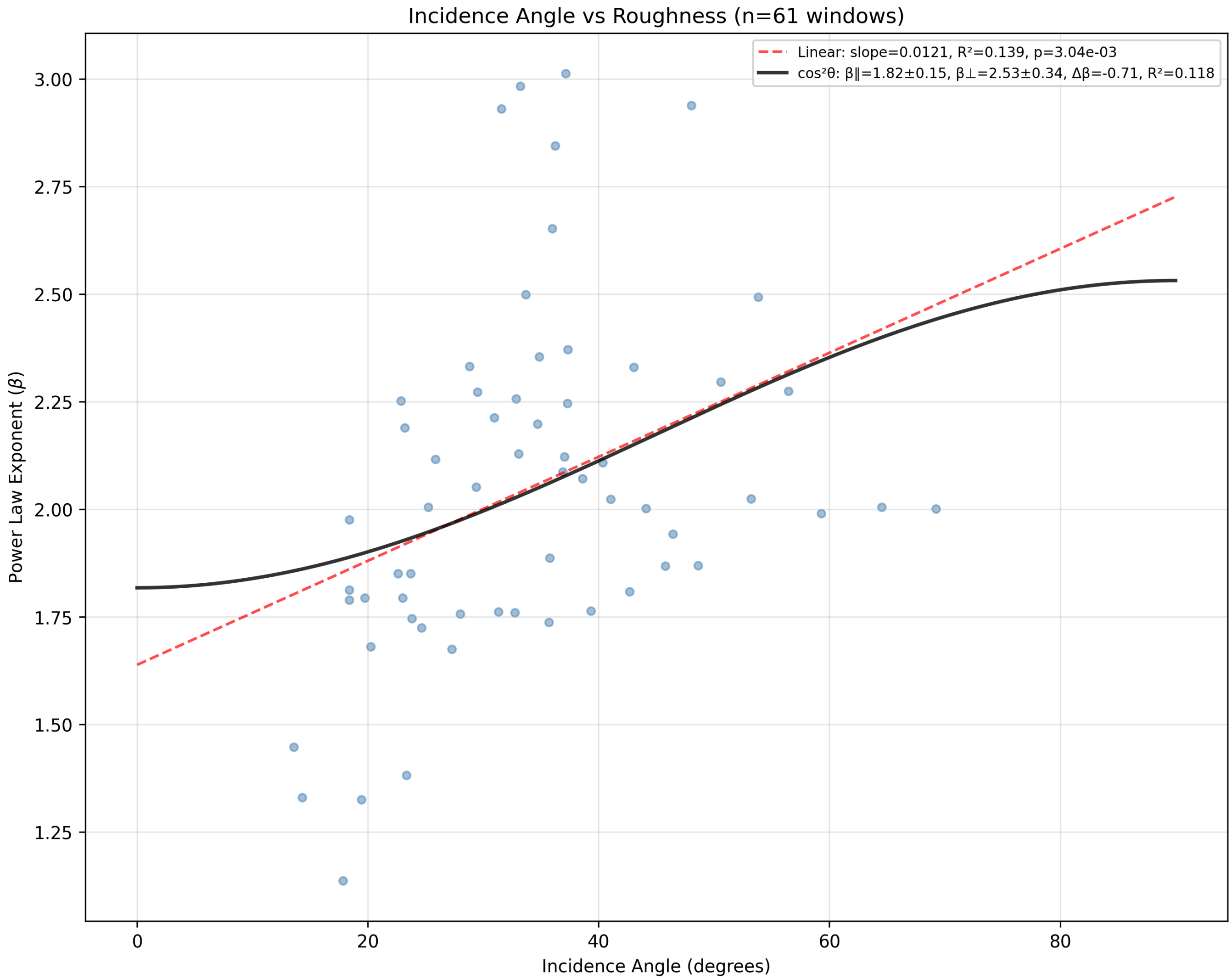
The raw max-minus-min elevation value for the vertical relief of 2194.1 m (single trajectory)

The height difference between the highest and lowest points available in the data for this area are Min: -1073.1 m | Max: 2912.3 m (across the single available trajectory).

The mean confirmed wavelength of 503.6 m with the range going up to ~33 km indicates the roughness is dominated by structures at roughly the sub-km-scale. These features are consistent with alpine subglacial terrain with well-developed valley systems. The higher mean wavelength (compared to hard bed regions) aligns with the $\beta = 2.3$ (the steeper spectrum concentrates power in longer wavelengths), and the erosion processes creating the topography operate at larger spatial scales.

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=0.0121, R^2=0.139, p=3.04e-03

Since **p < 0.05** means significant. At p=0.003 this result is significant (there's a 0.3% chance you'd see this slope (or steeper) from pure noise). The linear trend is real.

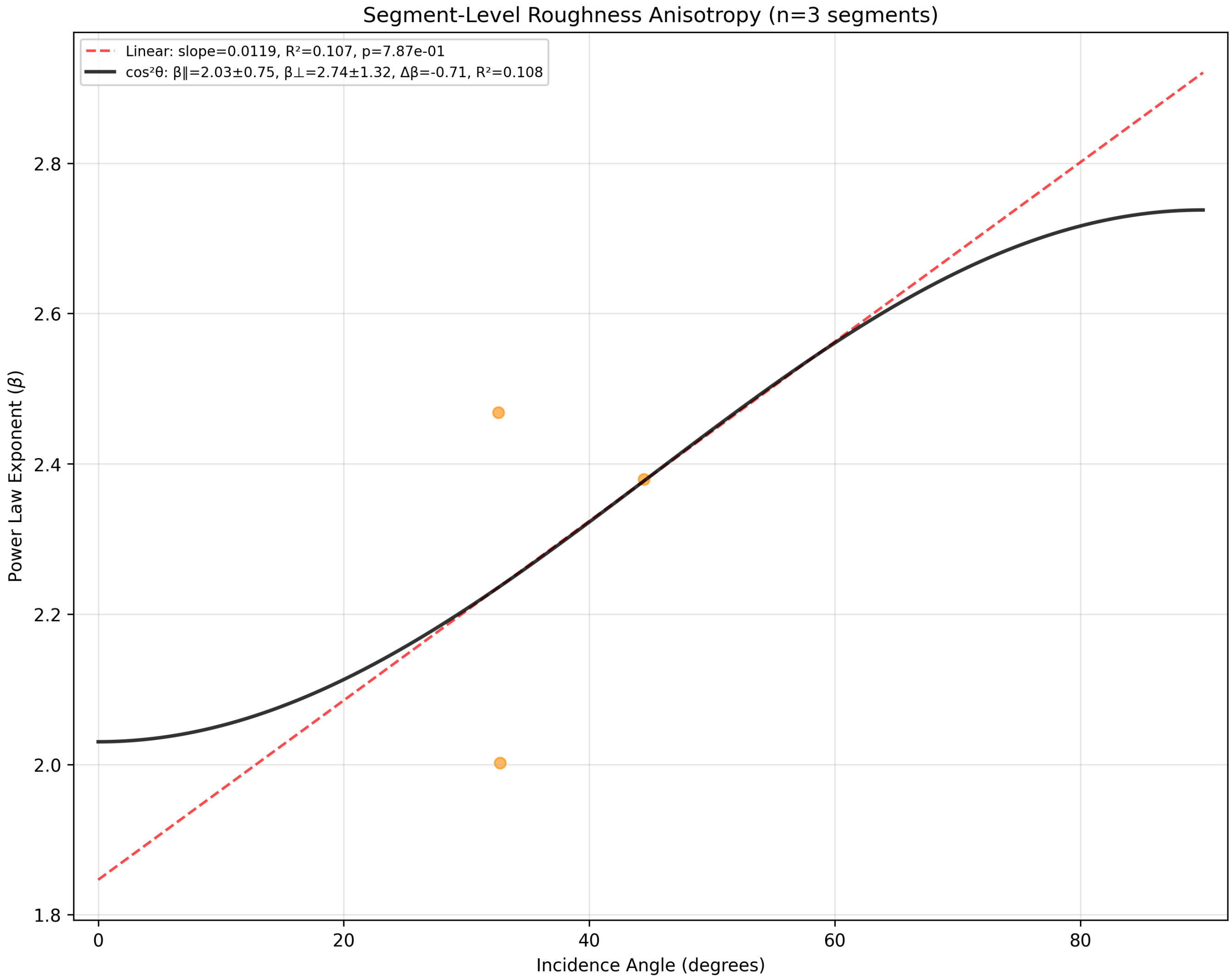
The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = -0.714$, which indicates a reversed anisotropy signature (roughness increases parallel to flow and decreases perpendicular to flow) This is likely indicating the landscape class is **alpine**, where flow-parallel transects cross ridges/valleys perpendicularly, encountering more roughness.

The fit z-scorecan be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2+\sigma_{\perp}^2}} = \frac{0.714}{\sqrt{0.154^2+0.337^2}} \approx \frac{0.714}{0.137} \approx 0.371$

This is below 1.96 (95% confidence interval), which means that the directionally

dependent β values are not significantly distinguishable.
Since the dataset R²=0.1181 implies that the model explains only 11% of variance.

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=0.0119, R²=0.107, p=7.87e-01
At p=0.787 this result is non-significant (there's a 79% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\|} - \beta_{\perp} = -0.708$, which provides support for the **Alpine landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\|}^2+\sigma_{\perp}^2}} = \frac{0.708}{\sqrt{0.753^2+1.322^2}} \approx \frac{0.708}{1.521} \approx 0.465$

This is below 1.96 (95% confidence interval), which means that the result is not significant. R²=0.1078 implies that the model explains 11% of variance. However the The dataset is too small to draw any meaningful conclusion from this analysis.

Conclusion:

The $\cos^2\theta$ two-parameter decomposition doesn't resolve the parallel vs perpendicular components well, likely because there is **only 3 segments and 61 windows along a single flight line** (the angular coverage may be too narrow or too correlated to separate the two terms)

Moller Stream

> ✎ **Hypothesis**
>
> Selective erosion (ice streams/ relict) / Alpine (subglacial) / Alpine (subaereal)

Terminal

```
=== Moller_Stream SUMMARY ===
------------------------------------------------------------
  RESULTS SUMMARY (64 trajectories aggregated)
------------------------------------------------------------
VERTICAL RELIEF (Max-Min):
  -> Mean: 1837.4m | Range: [488.2, 2921.2]m
ABSOLUTE BED ELEVATION:
  -> Min: -2115.2m | Max: 1818.0m  (across all trajectories)
AVG SEGMENT LENGTH:
  -> Mean: 215650.7m | Range: [36254.8, 636355.2]m
POWER LAW EXPONENT (Beta):
  -> Mean: 2.1 | Range: [1.3, 3.4]
HURST EXPONENT:
  -> Mean: 0.6 | Range: [0.2, 1.2]
MEAN ICE THICKNESS:
  -> Mean: 4177.5m | Range: [3081.4, 4789.3]m
............................................................
CONFIRMED WAVELENGTHS (Physically valid < L/2):
  -> Count: 6319
  -> Mean: 480.6m | Range: [30.4, 33970.2]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
```

```
  -> Count: 10
  -> Range: [6308m, 32975m]
............................................................
============================================================
Exported 1680 window rows and 115 segment rows
```

Observations

IFPA categorised the region containing these radar tracks as: Selective erosion / Alpine (subglacial) / Alpine (subaereal). The ODSA results indicate this region has a mean β value of 2.1 with a wide range: [1.3, 3.4]. Making the result a combination of **Hard bed** and **sedimentary** landscapes.

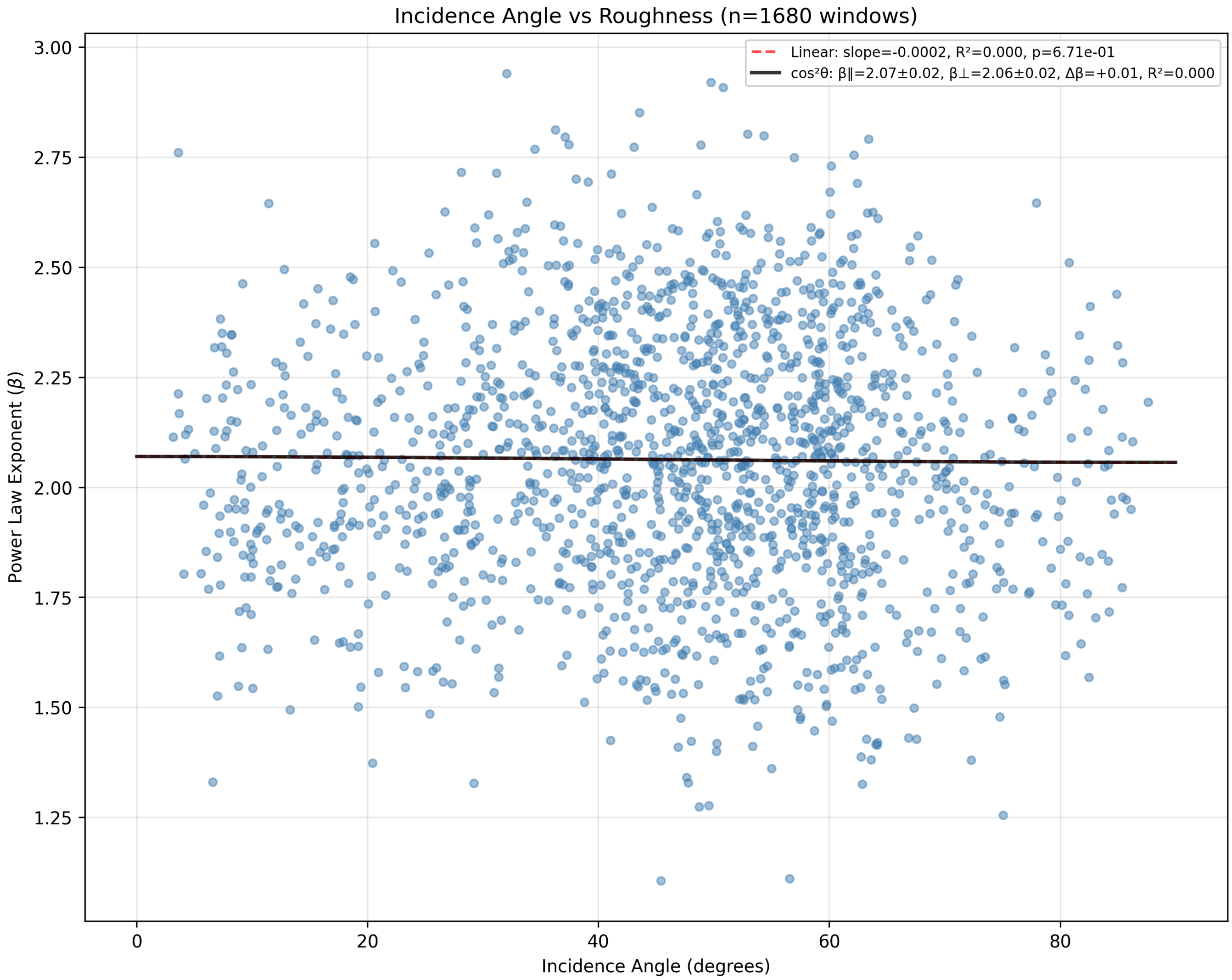
The mean vertical relief value is 1837.4 m.

The height difference between the highest and lowest points available in the data for this area are Min: -2115.2 m | Max: 1818.0 m (across all available trajectories).

The mean confirmed wavelength of 480.6 m with the range going up to ~34 km indicates the roughness is dominated by structures at roughly the sub-km-scale. This dataset is very rich in features with a huge confirmed wavelength count (6,319 peaks across 64 trajectories). The 10 candidate wavelengths ranging from 6--33 km are
interesting they could correspond to the spacing of ice stream troughs and inter-stream ridges. The wide wavelength range [30 m -- 34 km] spanning three orders of magnitude reflects spatial heterogeneity (different trajectories are sampling fundamentally different terrain).

Anisotropy

Window-level Roughness FIT



$\cos^2\theta$ fit: $\beta\|$=2.070±, $\beta\perp$=2.056±, $\Delta\beta$=, R^2=

The window level analysis shows a linear fit with slope=-0.0002, R^2=0.000, p=6.71e-01

At p=0.671 this result is non-significant. This means that the slope is essentially zero.

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_\| - \beta_\perp = +0.013$, which provides support for the **Alpine landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_\|^2+\sigma_\perp^2}} = \frac{0.013}{\sqrt{0.022^2+0.022^2}} \approx \frac{0.013}{0.0311} \approx 0.42$

This is below 1.96 (95% confidence interval), which means that the result is not significant. Since R^2=0.0001 implies that the model explains only 0.01% of variance.This means that **essentially no directional dependence is detected by the model.**

Conclusion:

Model shows no systematic dependence of roughness on flow orientation.

Thwaites BAS

Hypothesis

Isotropy ($\beta_{\parallel} = \beta_{\perp}$) or **anisotropy** ($\beta_{\parallel} > \beta_{\perp}$).
High relief
Alpine (subglacial/ subaereal) / Selective erosion (ice streams)

Terminal

```
=== Thwaites_BAS SUMMARY ===
------------------------------------------------------------
  RESULTS SUMMARY (12 trajectories aggregated)
------------------------------------------------------------
VERTICAL RELIEF (Max-Min):
  -> Mean: 664.8m | Range: [16.0, 1537.6]m
ABSOLUTE BED ELEVATION:
  -> Min: -1270.4m | Max: 2379.9m  (across all trajectories)
AVG SEGMENT LENGTH:
  -> Mean: 86259.1m | Range: [12518.9, 143710.1]m
POWER LAW EXPONENT (Beta):
  -> Mean: 2.3 | Range: [1.2, 3.5]
HURST EXPONENT:
  -> Mean: 0.7 | Range: [0.1, 1.3]
MEAN ICE THICKNESS:
  -> Mean: 523.7m | Range: [6.7, 803.8]m
............................................................
CONFIRMED WAVELENGTHS (Physically valid < L/2):
  -> Count: 1792
  -> Mean: 813.5m | Range: [30.4, 28845.3]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
  -> Count: 6
  -> Range: [6031m, 24677m]
............................................................
============================================================
```

```
Exported 269 window rows and 47 segment rows
```

Observations

IFPA categorised the region containing these radar tracks as **Alpine (subglacial) / Selective erosion (relict)**. The ODSA results indicate this region has a mean β value of 2.3 with a broad range: [1.2, 3.5]. This result is indicative of a heterogeneous landscape including **Hard bed** and **sedimentary** regions.

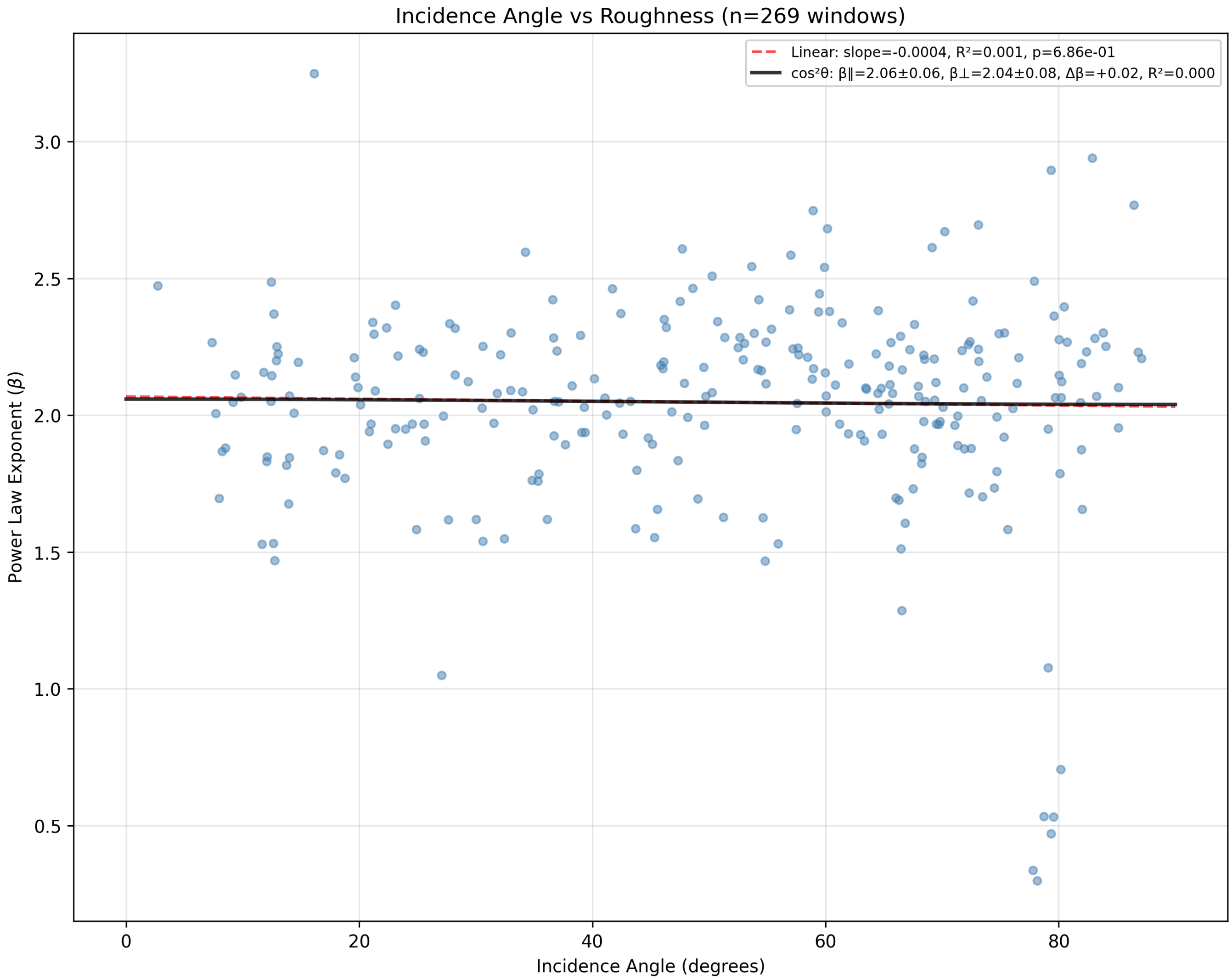
The mean value for the vertical relief is 664.8 m

The height difference between the highest and lowest points available in the data for this area are Min: -1270.4 m | Max: 2379.9 m (across all available trajectories).

The mean confirmed wavelength of 813.5 m with the range going up to ~29 km indicates the roughness is dominated by structures at roughly the sub-km-scale. With mean ice thickness of only 524 m, the **dominant roughness wavelength exceeds the ice thickness**. This matters because **roughness at wavelengths larger than ice thickness affects basal drag differently than sub-thickness roughness** (the ice
can conform to longer wavelengths).
The 6 candidate wavelengths at 6--25 km could correspond to major topographic features in the Thwaites catchment.
The broad beta range [1.2, 3.5] suggests different parts of the region have very different spectral slopes, and the higher mean wavelength may reflect the transition from crystalline bedrock inland to sediment-draped areas near the grounding line.

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=-0.0004, R^2=0.001, p=6.86e-01

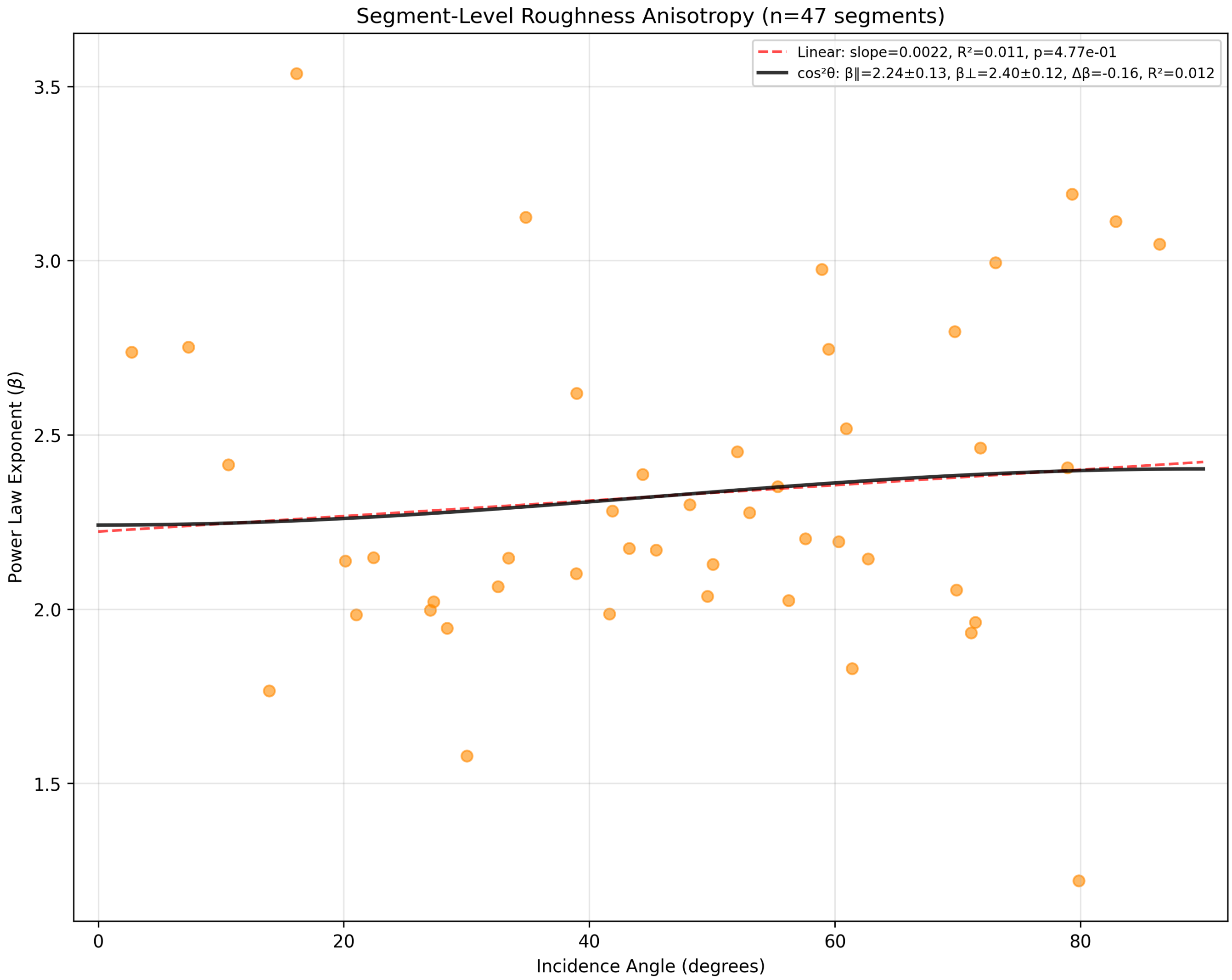
At p=0.69 this result is non-significant. This validates that the slope is essentially zero.

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.020$, which provides support for the **Alpine landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2+\sigma_{\perp}^2}} = \frac{0.020}{\sqrt{0.058^2+0.084^2}} \approx \frac{0.020}{0.1021} \approx 0.196$

This is below 1.96 (95% confidence interval), which means that the result is not significant. Since the dataset is small and the R^2=0.0003 implies that the model explains only 0.03% of variance.This means that **essentially no directional dependence is detected by the model.** This provides support for the **alpine** landscape classification (isotropic).

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=0.0022, R^2=0.011, p=4.77e-01
At p=0.477 this result is non-significant (there's a 47% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\|} - \beta_{\perp} = -0.161$, which provides support for the **Alpine landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\|}^2+\sigma_{\perp}^2}} = \frac{0.161}{\sqrt{0.135^2+0.120^2}} \approx \frac{0.161}{0.181} \approx 0.891$

This is below 1.96 (95% confidence interval), which means that the result is not significant. R^2=0.0122 implies that the model explains 1.22% of variance.

Conclusion:

Model shows no systematic dependence of roughness on flow orientation.

Thwaites CRESIS

Hypothesis

Isotropy ($\beta_{\parallel} = \beta_{\perp}$) or ($\beta_{\parallel} > \beta_{\perp}$)
High relief
Alpine (subglacial) / Selective erosion (relict)

Terminal

```
=== Thwaites_CR SUMMARY ===
------------------------------------------------------------
  RESULTS SUMMARY (10 trajectories aggregated)
------------------------------------------------------------
VERTICAL RELIEF (Max-Min):
  -> Mean: 178.1m | Range: [94.9, 328.3]m
ABSOLUTE BED ELEVATION:
  -> Min: -1641.3m | Max: -896.3m  (across all trajectories)
AVG SEGMENT LENGTH:
  -> Mean: 2167409.1m | Range: [51998.0, 12089012.1]m
POWER LAW EXPONENT (Beta):
  -> Mean: 0.2 | Range: [-1.0, 1.7]
HURST EXPONENT:
  -> Mean: -0.4 | Range: [-1.0, 0.3]
MEAN ICE THICKNESS:
  -> Mean: 1269.2m | Range: [1155.7, 1441.6]m
............................................................
CONFIRMED WAVELENGTHS (Physically valid < L/2):
  -> Count: 17548
  -> Mean: 1347.1m | Range: [106.4, 47828.4]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
  -> Count: 33
  -> Range: [7412m, 48800m]
............................................................
============================================================
Exported 49064 window rows and 805 segment rows
```

Observations

IFPA categorised the region containing these radar tracks as Alpine (subglacial) / Selective erosion (relict). The ODSA results indicate this region has a mean β value of 0.2 with a range: [-1.0, 0.3]. **Making the result anomalous.** ==This is almost certainly a data/processing artifact rather than a geological signal. ==

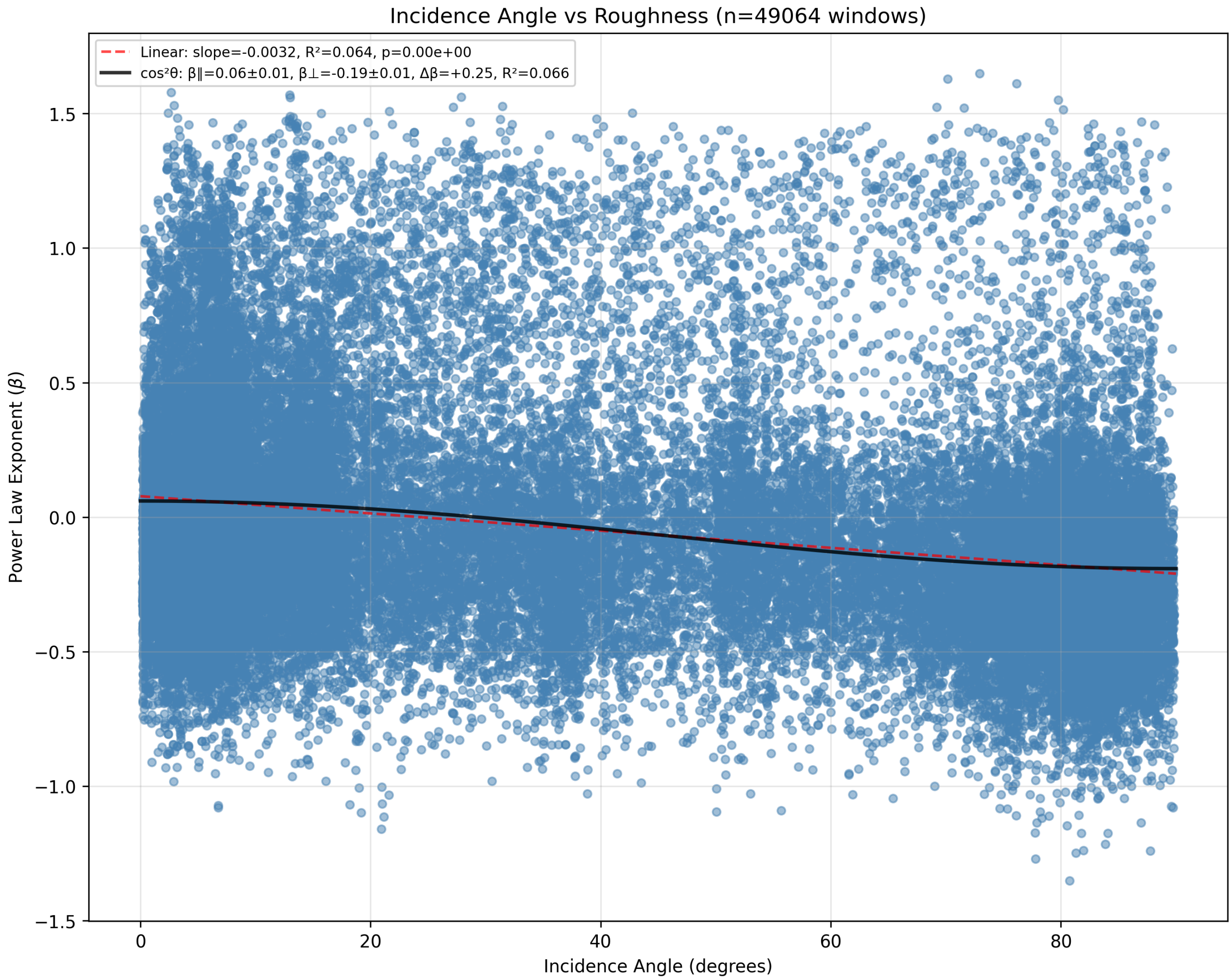
The mean vertical relief of 178.1 m.

The height difference between the highest and lowest points available in the data for this area are Min: -1641.3 m | Max: -896.3 m.

The mean confirmed wavelength of 1347.1 m with the range going up to ~48 km indicates the roughness is dominated by structures at roughly the km-scale. Because the segments are extremely long, the Lomb-Scargle periodogram fits across a massive frequency range. The 33 candidate wavelengths reaching up to 49 km suggest the periodogram is picking up structure at scales approaching the window size itself. The minimum confirmed wavelength of 106 m (vs 30 m for other regions) is also suspicious (implying the short-wavelength peaks are being suppressed, likely because the near-flat spectrum (beta~0.2) means there's no significant power concentration at any particular scale. These are likely the same data artifacts affecting β.

Anisotropy

Window-level Roughness FIT



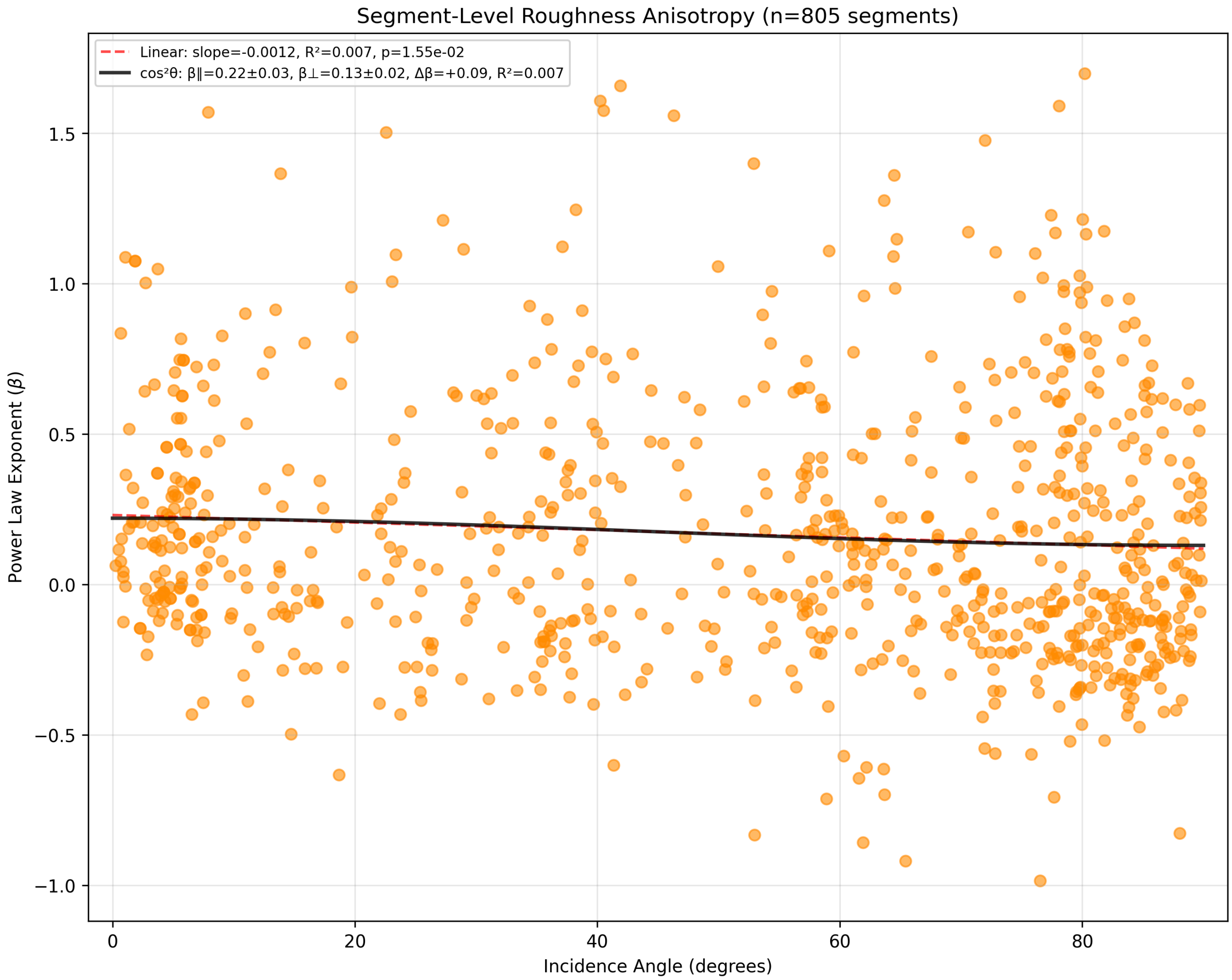
The window level analysis shows a linear fit with slope=-0.0032, R^2=0.064, p=0.00e+00
Since **$p < 0.05$** means significant.

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.251$, which provides support for the **Alpine landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2+\sigma_{\perp}^2}} = \frac{0.251}{\sqrt{0.005^2+0.005^2}} \approx \frac{0.251}{0.007} \approx 35.5$

This is over 1.96 (95% confidence interval), which means that the result is significant. Since R^2=0.0662 implies that the model explains only 6.62% of variance. This means that **essentially there is directional dependence detected by the model.** This provides support for the **selective erosion** class.

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=-0.0012, R^2=0.007, p=1.55e-02
At p=0.0155 this result is significant (there's a 1.55% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_\parallel - \beta_\perp = +0.090$, which provides support for the **Selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_\parallel^2+\sigma_\perp^2}} = \frac{0.090}{\sqrt{0.028^2+0.022^2}} \approx \frac{0.090}{0.0356} \approx 2.528$

This is larger than 1.96 (95% confidence interval), which means that the result is statistically significant. However the R^2=0.0066 value implies that the model explains 0.66% of variance.

NOTES

The CRESIS dataset has extremely long average segment lengths (mean ~2167 km, max ~12,000 km), an order of magnitude longer than any other region. At these scales, the Lomb-Scargle periodogram fits power laws across an enormous frequency range where spectral leakage and red-noise contamination dominate. The near-zero β means the PSD is effectively flat (equal power at all frequencies), characteristic of instrument noise rather than real topography. The negative Hurst exponent ($\zeta = -0.4$) further confirms this, as a negative ζ is physically meaningless for a self-affine surface. The CRESIS β values should not be interpreted glaciologically.

Conclusion:

Anomalous (mean $\beta \approx 0.2$). The anisotropy result matches that of Selective erosion (relict) consistent with IFPA's classification. **The relief data (mean 178 m, all elevations below sea level (-1641 to -896 m) reliable and indicate a low-relief, deeply submerged basin**

For now I will consider the results for this region as invalid: will consider further analysis with a segment length cap (~300km?)

Dronning Maud Land (DML)

Hypothesis

High anisotropy ($\beta_{\|} > \beta_{\perp}$)
Selective erosion (ice streams) / invalid data

Terminal

```
=== DML_AniRES SUMMARY ===
------------------------------------------------------------
  RESULTS SUMMARY (8 trajectories aggregated)
------------------------------------------------------------
VERTICAL RELIEF (Max-Min):
  -> Mean: 447.5m | Range: [153.1, 918.4]m
ABSOLUTE BED ELEVATION:
  -> Min: -951.6m | Max: 534.2m  (across all trajectories)
```

```
AVG SEGMENT LENGTH:
  -> Mean: 150948.9m | Range: [31847.9, 284789.8]m
POWER LAW EXPONENT (Beta):
  -> Mean: 2.3 | Range: [1.6, 3.3]
HURST EXPONENT:
  -> Mean: 0.6 | Range: [0.3, 1.1]
MEAN ICE THICKNESS:
  -> Mean: 286.3m | Range: [192.8, 417.6]m
............................................................
CONFIRMED WAVELENGTHS (Physically valid < L/2):
  -> Count: 1587
  -> Mean: 659.0m | Range: [67.8, 31050.3]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
  -> Count: 3
  -> Range: [6390m, 34086m]
............................................................
============================================================
Exported 316 window rows and 36 segment rows
```

Observations

IFPA categorised the region containing these radar tracks as Selective erosion (ice streams) / invalid data . The ODSA results indicate this region has a mean β value of 2.3 with a range: [1.6, 3.3]. Putting the result in a combination landscape classification between **hard bed** and **sedimentary**.

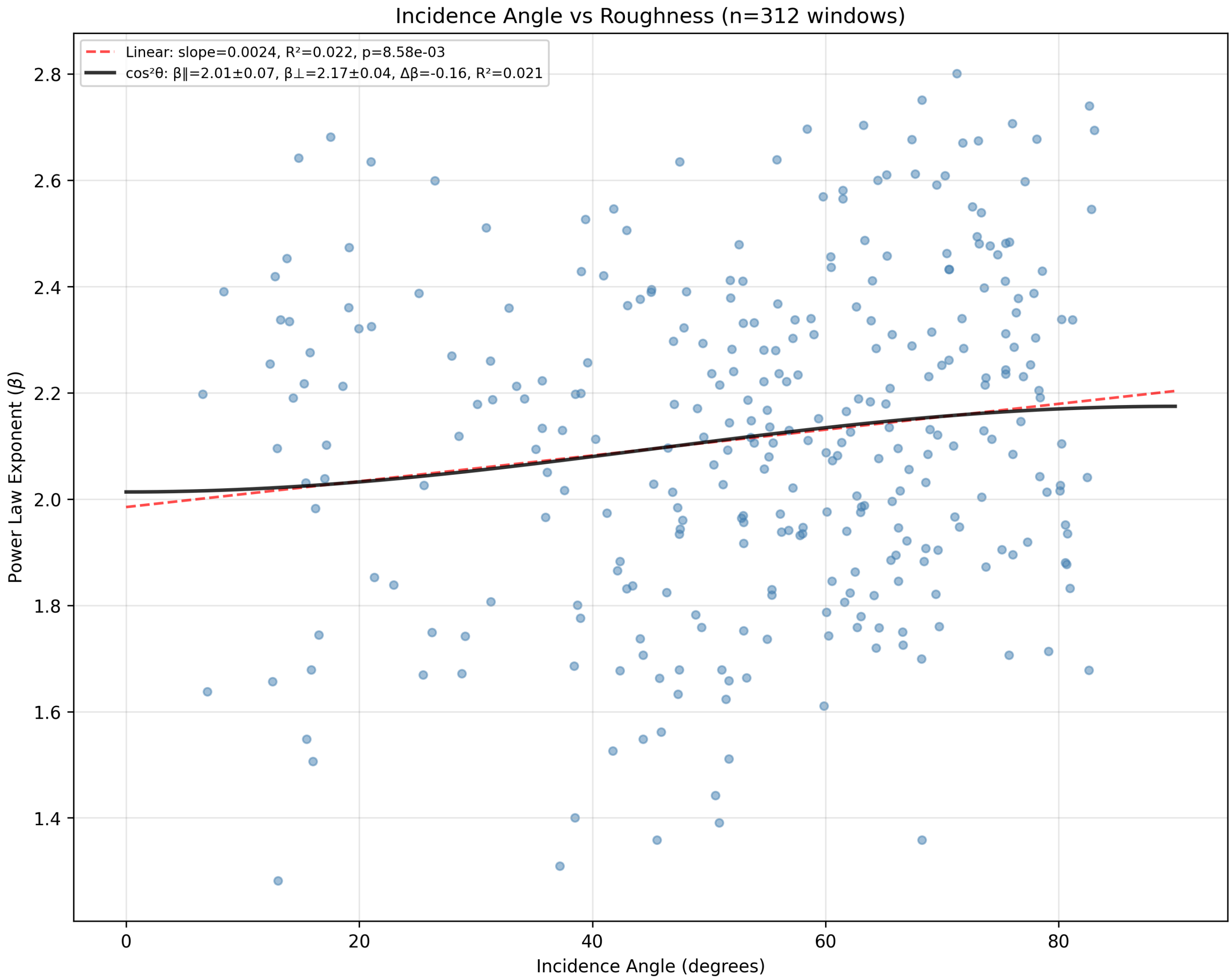
The mean the vertical relief of 447.5 m with a range: [153.1, 918.4]m.

The height difference between the highest and lowest points available in the data for this area are Min: -951.6 m | Max: 534.2 m (across all trajectories).
The mean confirmed wavelength of 659.0 m with the range going up to ~ 31 km indicates the roughness is dominated by structures at roughly the sub-km-scale. Thinnest ice region with mean thickness 286 m. The dominant roughness wavelength is more than double the ice thickness, which is significant for ice dynamics (the bed roughness is at scales the ice sheet cannot conform to, implying strong form drag. The 3 candidate wavelengths at 6--34 km could correspond to major geological structures.
The beta of 2.3 and wavelength of 659 m together suggest an erosional landscape

with well-developed features at the sub-km to km scale, consistent with the IFPA classification of selective erosion (ice streams).

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=0.0024, R^2=0.022, p=8.58e-03

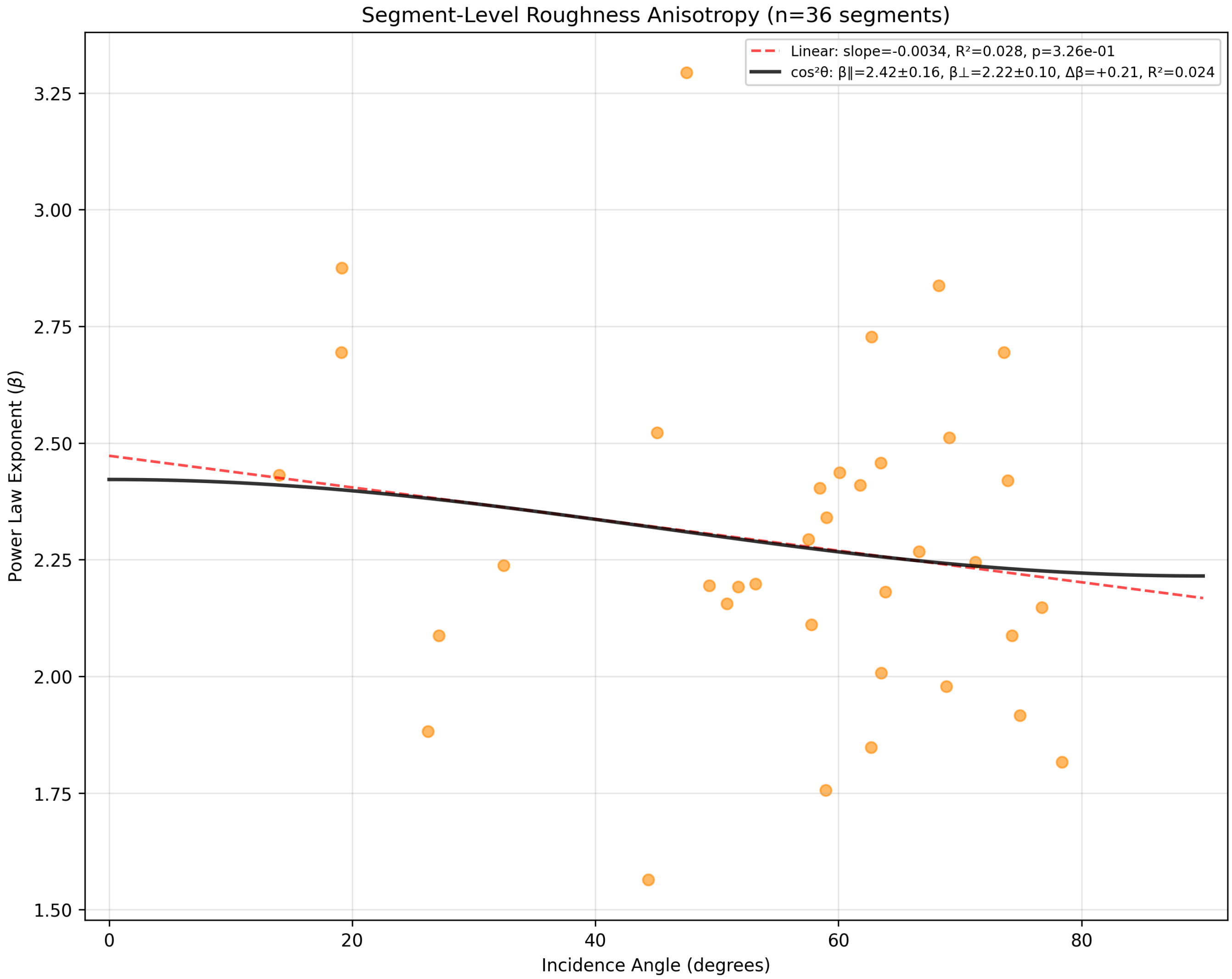
At p=0.00858 this result is significant (there's a 0.86% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\|} - \beta_{\perp} = -0.161$, which provides support for the **Alpine landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\|}^2+\sigma_{\perp}^2}} = \frac{0.161}{\sqrt{0.069^2+0.044^2}} \approx \frac{0.161}{0.0818} \approx 1.97$

This is just above 1.96 (95% confidence interval), which means that the result is significant but practically negligible, since $R^2=0.0209$ this implies that the model explains 2.1% of variance.

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=-0.0034, $R^2=0.028$, p=3.26e-01
At p=0.326 this result is not significant (there's a 32% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_\parallel - \beta_\perp = +0.207$, which provides support for the **Selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_\parallel^2+\sigma_\perp^2}} = \frac{0.207}{\sqrt{0.162^2+0.095^2}} \approx \frac{0.207}{0.188} \approx 1.102$

This is smaller than 1.96 (95% confidence interval), which means that the result is statistically significant. In addition the $R^2=0.0241$ value implies that the model explains 2.4% of variance, which indicates a pretty weak fit to the data.

Conclusion:

Directional dependence is detected by the model at the window level (indicates a reversed anisotropy signal - Alpine?). Conversely at the segment level the analysis indicates a selective erosion signature. However roughness variance is barely explained by the fits.